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End-to-End E-commerce Intelligence System: Building a Customer 360 Analytics Framework

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Business Problem Statement

Modern e-commerce platforms generate data across multiple systems, including customers, orders, payments, products, sellers, and reviews. These datasets are fragmented and require integration to derive meaningful business insights. The objective of this project is to integrate multiple datasets into a unified Customer 360 analytics framework and analyze customer behavior, revenue patterns, seller performance, operational efficiency, and customer satisfaction.

Dataset Description

- customers.csv – Customer details and location information
- orders.csv – Order details and purchase date
- order_items.csv – Product-level order details
- payments.csv – Payment information
- reviews.csv – Customer review and satisfaction data
- products.csv – Product details and categories

- sellers.csv – Seller information
- geolocation.csv – Geographic information
- category_translation.csv – Product category translation mapping

Methodology

- Data Loading and Initial Exploration using Pandas
- Data Cleaning and Preprocessing
- Handling Missing Values and Removing Duplicates
- Datetime Conversion and Data Type Validation
- Data Integration using Merge Operations
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Data Visualization using Matplotlib and Seaborn
- Business Insight Generation and Recommendations

Feature Engineering

- Total Order Value
- Delivery Time
- Items per Order
- Customer Purchase Frequency
- Customer Lifetime Value
- Average Order Value per Customer

Exploratory Data Analysis

- Customer Analysis
- Revenue and Order Analysis
- Product Analysis
- Seller Analysis
- Review and Satisfaction Analysis

Key Findings

- Most customers are one-time buyers, while repeat customers contribute higher revenue.
- Certain product categories generate the majority of sales and revenue.
- Delivery delays negatively impact customer review scores.
- A small number of sellers contribute significantly to overall revenue.
- Monthly sales trends show seasonal fluctuations and peak sales periods.

Business Insights

- Customer retention is critical for long-term business growth.
- Efficient logistics operations improve customer satisfaction.
- Top-performing product categories should receive increased marketing focus.
- Seller performance monitoring can improve operational efficiency.
- Geographic customer distribution supports targeted marketing strategies.

Recommendations

- Implement loyalty programs to improve repeat customer purchases.
- Reduce delivery delays through logistics optimization.
- Promote high-performing product categories strategically.
- Support low-performing sellers with operational guidance.
- Use customer behavior analysis for personalized marketing campaigns.

Conclusion

The project successfully integrated multiple e-commerce datasets into a unified analytical framework. Using preprocessing, feature engineering, EDA, and visualization techniques, meaningful business insights were generated regarding customer behavior, revenue patterns, operational performance, and customer satisfaction.